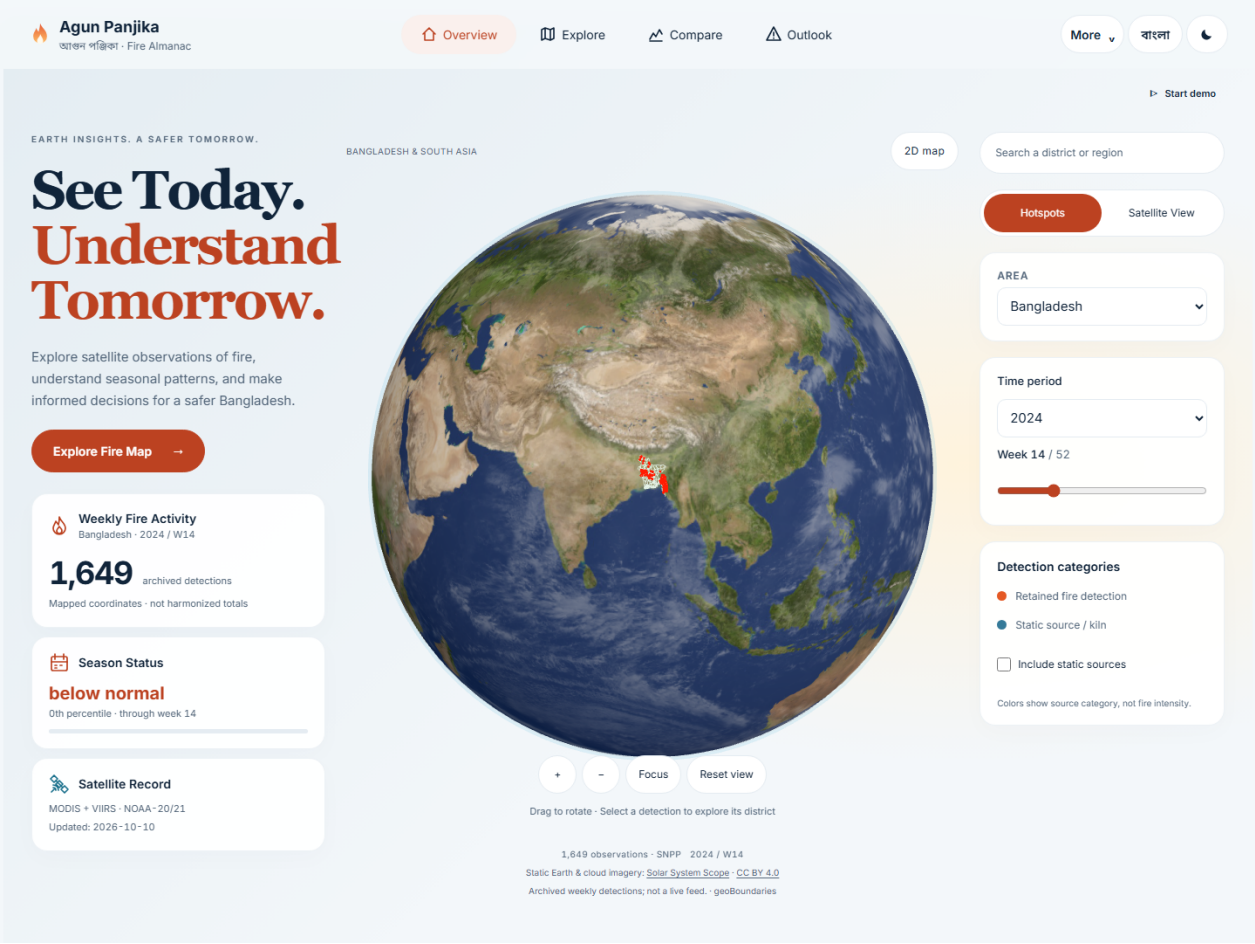
NASA SPACE APPS CHALLENGE 2026

Agun Panjika

আগুন পঞ্জিকা

Project brief for academic review

A comparable fire history. A clearer view of the season.



Team KUETaguns

Khulna University of Engineering & Technology (KUET), Bangladesh

Harmonization of MODIS and VIIRS Hot Spots

Prepared for mentorship and research discussion
Reviewed 29 October 2026.
Data snapshot: 9 October 2026.

Source code and documentation
Project video
Global extension proposal

1 Executive overview

The question the project helps a user investigate

Is this fire season normal for this area?

Agun Panjika (Fire Almanac) is an English/Bangla web application and a Python pipeline for comparing Bangladesh's satellite-detected fire activity across changes in observing instruments. It starts with Bangladesh, particularly the Chittagong Hill Tracts, and is intended for forest officers, communities, researchers and newsrooms.

Why harmonization is necessary

MODIS and VIIRS observe at different spatial resolutions and sensitivities. NASA FIRMS provides MODIS active-fire layers at approximately 1 km and VIIRS at 375 m. Joining their raw counts can make an instrument change look like a change in burning. Those resolutions describe observations, not the size or number of unique fires.

The pipeline reconstructs earlier weekly activity on a VIIRS S-NPP detection scale, distinguishes observed and estimated periods, and exposes uncertainty. The interface connects that comparable history to calendars, maps, seasonal normals and shareable reports.

2001 onward
Historical record, with early periods reconstructed

64 districts
Five analysis regions across Bangladesh

25.25%
Lower internal weekly MAE than a fitted factor

What the current evidence supports

The committed internal leave-one-year-out evaluation for 2012–2025 reports national weekly mean absolute error (MAE) of 36.964 detections, versus 49.450 for a single factor fitted within each training fold. This is a promising reconstruction benchmark, not an independently confirmed final result or operational forecast accuracy.

The software is a working prototype. The 29 October checks passed 59 Python tests, typechecking, the production build and eleven browser test groups covering 42 route/theme/language/viewport views. A field-user study, prospective warning evaluation and reliable hosted refresh have not been established.

Purpose of academic mentorship

The immediate need is to strengthen evaluation independence, position the contribution against prior work, and decide whether the next milestone should be a deeper Bangladesh user pilot or an independently evaluated transfer case.

Reading guide

Data and product: page 3. Method and architecture: page 4. Internal results: page 5. Continuity and outlooks: page 6. Verification and limitations: page 7. Research questions and evidence links: page 8.

2 Data, geography and implemented product

Data provenance

Source	Role in the project	Interpretation limit
NASA FIRMS	MODIS Terra/Aqua; VIIRS S-NPP, NOAA-20 and NOAA-21 observations	A detection is not a unique fire; omissions and repeat observations remain possible.
NASA POWER	Historical weather, FWI inputs and Dhaka aerosol context	Regional sampled/reanalysis context, not local station observations.
Open-Meteo	Forecast weather for experimental danger views	Actual forecast-feed skill needs prospective evaluation.
geoBoundaries	District/upazila assignment and administrative maps	Boundary versions and snapping influence local interpretation.
ESA Fire_cci	Burned-area cross-check: MRBA v6.0, then Sentinel-3 SYN v1.1	Different observable and a product change in 2019.
NOAA CPC ONI	Experimental ENSO covariate	Inclusion in an experiment does not establish improved skill.

Table 1. Data sources and their roles. Full attribution is linked on page 8.

What is in the saved snapshot?

The national record contains 1,345 ISO weeks from 2001-W01 through 2026-W41: 576 estimated and 769 observed. No successor-bridged week is present in that saved record. Observations end on **9 October 2026**; exports were generated on 10 October. The historical-normal baseline is 2003–2025. The saved weather forecast ends on 25 October, so the outlooks are archived examples as of this review.

The five analysis regions are Rangamati, Khagrachhari, Bandarban, Chattogram/Cox's Bazar hills, and the rest of Bangladesh. Local calendars cover 25 upazilas and 124 unions in the three CHT districts: Rangamati 10/51, Khagrachhari 8/41 and Bandarban 7/32. Earlier local values are apportioned estimates, not independently fitted local forecasts. Union names still need full Bangla localization and community review.

Implemented user experience

The seven routes provide an overview, fire calendar, sensor comparisons, map, experimental outlook, cross-border context and methodology. Users can inspect raw/factor/harmonized series, uncertainty, district patterns, archived point coordinates and seasonal percentiles; export CSV/PNG/print reports; or follow a bilingual guided demonstration.

The restored **Soft Earth** homepage uses a textured Three.js globe with linked area/year/week controls and a 2D fallback. The Earth/cloud textures are static geographic context. Shared URL state, mobile layouts, keyboard/table alternatives, loading/error/retry handling and cached offline access are implemented. External map tiles/fonts and unvisited resources are outside the offline guarantee.

3 Method and system design

From observations to comparable weekly history

The pipeline normalizes FIRMS metadata, removes exact duplicates, tags geography and filters confidence/type. It retains MODIS confidence at least 30% and VIIRS nominal/high observations; excludes FIRMS static-land/offshore types; and uses persistence, day/night and seasonality to flag probable industrial heat cells. District-week counts, radiative power and sensor-composition features are aggregated to regions and the nation.

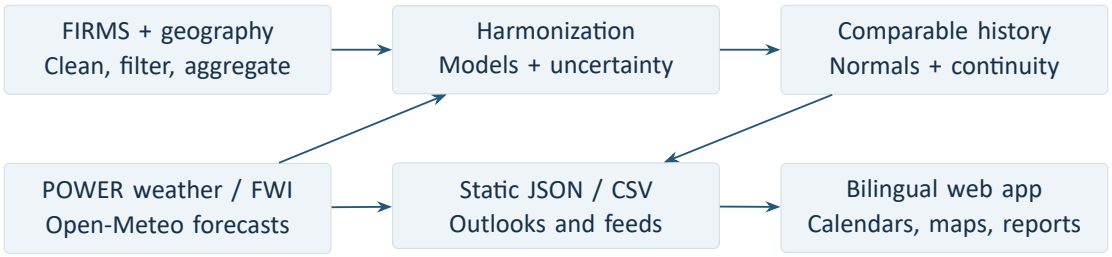


Figure 1. Implemented data flow. ESA burned area supplies a separate external cross-check.

Reconstruction model

During the 2012–2025 overlap, regional weekly S-NPP counts are the target. A negative-binomial model uses MODIS count/FRP with regional and seasonal intercepts. Bagged Poisson LightGBM adds sensor composition, neighbouring-week counts, cloud/weather/FWI and viewing geometry. The shipped point estimate is

$$\widehat{V}_{r,w} = 0.30\,\widehat{V}_{\mathrm{NB},r,w} + 0.70\,\frac{1}{5}\sum_{s=1}^{5}\widehat{V}_{\mathrm{GBM},s,r,w}. \tag{1}$$

Here r is region, w is week and s indexes five model seeds. A separate, lower-confidence Terra-only reconstruction covers 2001 through 2002-W26. Some features include **next-week MODIS counts**: this is retrospective reconstruction, not an issue-time prediction algorithm.

Uncertainty, continuity and season status

Nominal 80% intervals use held-out log-error distributions by activity/season, with separate annual-error treatment. Strictly nested calibration has not yet been established. Successor comparisons evaluate identity, fitted-factor and shrunk monthly corrections; both currently select identity. Season status ranks cumulative activity against complete historical years, excluding the displayed year.

Python prepares static exports; React/TypeScript/Vite consumes them without an application backend or database. D3 powers charts and the Canvas fallback, Three.js the textured Earth, and MapLibre the map. The pipeline/model code and exported evidence remain available for inspection.

4 Internal reconstruction results

The committed evaluation leaves out one overlap year at a time and compares national weekly predictions with S-NPP. The values below are means across held-out years over 2012–2025; no models were retrained for this report.

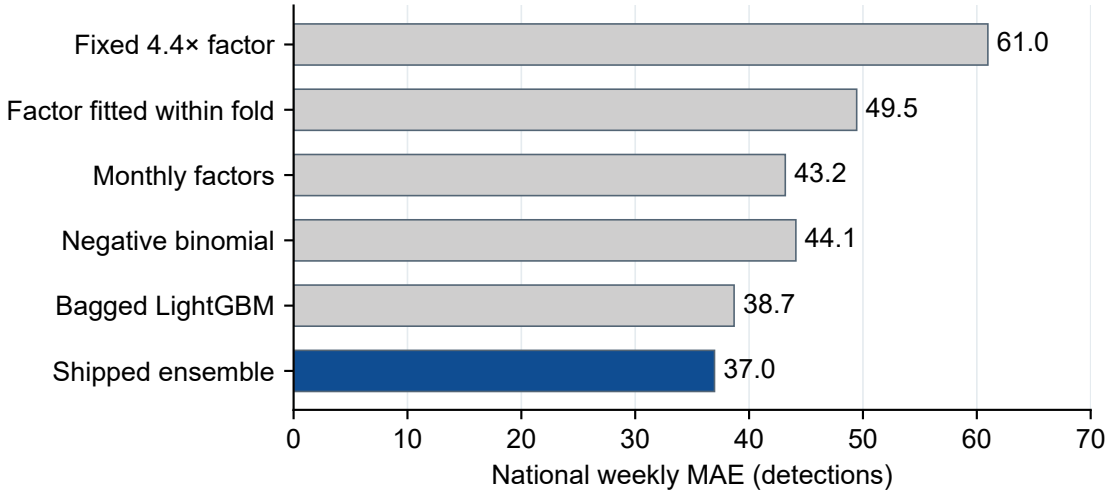


Figure 2. Internal national weekly MAE. Lower is better. Aggregate bars have no newly estimated confidence intervals.

Method	Weekly MAE	Off-season MAE	Annual error
Fixed 4.4× factor	60.971	26.095	20.150%
Factor fitted within fold	49.450	28.705	12.785%
Monthly factors	43.180	21.933	11.888%
Negative binomial	44.110	19.458	9.652%
Bagged LightGBM	38.684	**15.347**	10.426%
Shipped ensemble	**36.964**	15.861	**9.501%**

Table 2. National errors from the saved validation export. Annual error is mean absolute annual-total percentage error; weekly MAE is measured in detections.

The ensemble reduces weekly MAE by **25.25%** against the fitted factor and 39.37% against fixed 4.4×. It beats the fitted factor in all 14 evaluated off-seasons. LightGBM alone has the lowest overall off-season MAE; the ensemble does not win every metric, region or year.

Uncertainty is material, not decorative

Reported coverage is 80.1% for region-weeks, 81.3% for national weeks and 85.7% for national years, against an 80% target. Median national-week interval width is 1.39 times the estimate. Coverage should be interpreted alongside the calibration-independence concern.

Evidence: validation.json. The Terra-only model reports weekly MAE 66.76 versus 99.58 for a Terra scaling factor, with lower confidence in the earliest reconstruction. These are internal results, not an untouched final evaluation.

5 Continuity checks and experimental outlooks

An independent observable

ESA burned area provides a plausibility check on annual/seasonal variability. The saved transition diagnostic is 0.95 for harmonized detections versus 3.09 for a raw stitched record; estimated-years log-annual correlation is 0.93 and seasonal-cycle correlation 0.98.

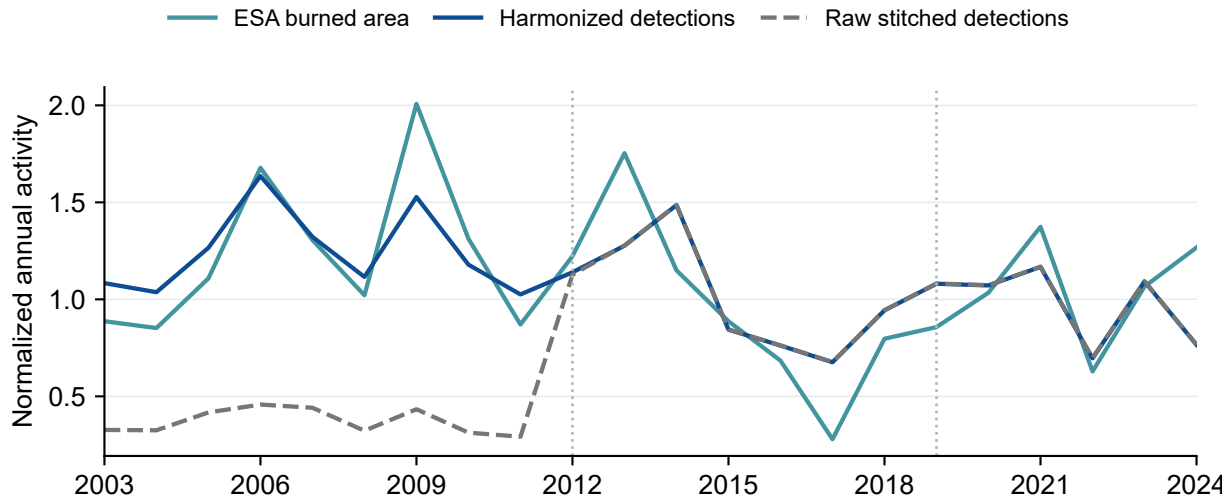


Figure 3. Annual series divided by their own 2012–2024 means. Markers identify VIIRS entry (2012) and the ESA product change (2019). Burned area and detection counts are different observables; agreement does not establish ground-truth fire-count accuracy.

Successor-satellite continuity

The selected identity bridge reports mean absolute held-out total errors of **4.58% for NOAA-20** and **10.79% for NOAA-21**. NOAA-21 has a shorter overlap including partial years. These results do not establish interchangeability in every district/season. Handover tests must cover main, local, neighbour and point products, missing observations and future nonidentity factors.

Two-week activity outlooks

Hindcast setting	Brier skill	ROC AUC
Fire history only	0.110	0.708
Fire history + observed future FWI	0.213	0.768

Table 3. Saved two-week hindcasts against climatology. The stronger weather experiment assumes access to future observed FWI.

The weather-assisted result is an optimistic sensitivity experiment, **not actual forecast-feed accuracy**. End-to-end temporal separation and independent forecast calibration remain unestablished. The remainder-of-season product reports no sufficient skill and shows the normal range. Cross-border/aerosol views are exploratory associations, not a causal smoke-arrival model.

Evidence: partner.json, bridge.json and forecast.json.

6 Verification, reproducibility and limitations

Check on 29 October	Observed result and scope
Python tests	59 passed; no skips in the available local environment.
TypeScript and production build	Passed for the restored Soft Earth interface.
Production browser regression	11 groups passed; 42 views with no recorded page/console errors, overflow or axe findings.
Visual restoration	Desktop screenshot matches the saved Soft Earth reference byte-for-byte; mobile view inspected.
Scientific regeneration	Not performed for this documentation; saved results were inspected.

Table 4. Software checks establish behavior under the tested setup. They are not field-user validation, full accessibility certification or a new scientific evaluation.

Research and release limits

1. **Independent evaluation.** Some calibration residuals come from models that can include the evaluation year in training. Model/feature/weight selection and all learned preprocessing need separation from an untouched final holdout.
2. **Coverage and issue-time availability.** Reconstruction uses following-week information; forecast hindcasts consume an already-reconstructed record. Missing acquisition coverage must be distinguished from genuine zero activity.
3. **Static filtering and small areas.** The tuned filter's final image sample has 16 positive industrial labels out of 20; it is not an independent precision/recall study. District/local apportionment does not establish small-area calibration.
4. **Users and operational impact.** The five-person protocol has uncompleted rows. Native Bangla/CHT review, uncertainty comprehension and workflow fit need actual users. No field adoption, fires prevented or operational warning authority is claimed.
5. **Release readiness.** A working public app URL was not verified; hosted refresh had failed. Stale-product handling, dependency advisories, explicit software licensing and in-app method-copy drift remain follow-up work.

Reproduce the prototype and inspect the science

Existing exports are enough to run the interface without a FIRMS key. Use `npm ci`, `npm run typecheck` and `npm run build` from `web/`. Python installation uses `python -m pip install -e ".[dev]"`; run `python -m pytest -q` from the root.

Full pipeline regeneration downloads inputs and fits models; inspect `python -m pipeline.run_all --help` first. Keep API credentials in the gitignored pipeline environment. Record data/configuration versions, seeds and output hashes for new experiments.

Instructions and provenance: verification report, machine-readable checks, validation notebook and source-level scientific review.

7 Research agenda and evidence index

Specific requests for academic guidance

1. **Evaluation independence.** Which temporal/geographic splits, nested calibration and locked final holdout would make the main harmonization result defensible?
2. **Contribution and related work.** Which prior sensor-fusion methods, strong baselines and ablations should frame this system's contribution? The project uses established models rather than claiming a new foundational algorithm.
3. **Transfer.** Should the next milestone be a deeper Bangladesh case or a validated second-country demonstration? Which contrasting setting would test the method meaningfully?
4. **Practical use.** Which forest/community decision should the interface support? A domain reviewer and practitioner introductions would help shape a credible pilot.
5. **Research potential.** After the missing experiments, is there a focused paper or technical-report question? Publishability is an open question, not an assumed outcome.

Claim-to-evidence check

Claim	Current assessment
Working bilingual prototype	Supported in the tested local environment by source, production captures and regression checks.
Lower reconstruction error	Supported as an internal benchmark; needs independent final confirmation.
Sensor continuity	Limited aggregate diagnostics support plausibility, not a bias-free or ground-truth record.
Guaranteed uncertainty, worldwide accuracy or operational impact	Not established; requires new evaluation and user evidence.

Selected sources and supporting materials

- NASA FIRMS data/layer documentation; NASA POWER; ESA Fire_cci; geoBoundaries; Open-Meteo.
- Farukh, Islam & Hayasaka (2023), Atmosphere 14(1):97: Bangladesh fire-season context, distinct from this project's validation.
- Full source/imagery credits and submission notes. Solar System Scope Earth/cloud textures are used under CC BY 4.0; basemap and imagery attributions remain in the app/repository.
- Detailed Markdown project brief; global extension plan; team video; source repository.

Proposed outcome of the first mentor discussion

Agree on one primary user journey, an independent evaluation protocol and the next bounded deliverable. The companion extension proposal gives milestones and acceptance criteria for that decision.